Juan José
Herrera Aranda



Personal Information

Full name: Juan José Herrera Aranda
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Nationality: Spanish
Birth date: 1999/07/29
Links: Personal website, Google Scholar, LinkedIn

Professional Summary

PhD candidate in Computer Science at the University of Granada, specialising in **zero-shot learning** and semantic-space preprocessing. My research experience encompasses machine learning, deep learning, generative artificial intelligence, and quantum machine learning, supported by a strong interdisciplinary background in mathematics and computer science. I have also conducted research at BBVA and as a visiting PhD student at the University of Nottingham.

Education

2023 – present
PhD in Computer Science
Thesis: "Preprocessing the Semantic Space in Zero-Shot Learning", University of Granada, Granada, Spain.
- Supervisors: Isaac Triguero Velázquez, Guillermo Gómez-Trenado, and Francisco Herrera Triguero.

2024 – 2026
Master's Degree in Data Science and Computer Engineering
University of Granada, Granada, Spain.
- Final grade: 9.26/10; GPA: 3.063/4.
- Master's thesis: "Semantic-Inductive Attribute Selection for Zero-Shot Learning."

2022 – 2023
Master's Degree in Quantum Computing
University of Antonio de Nebrija, Madrid, Spain.
- Master's thesis: "Quantum Generative Adversarial Networks."

2017 – 2022
Bachelor's Degree in Computer Science
University of Granada, Granada, Spain.
- Specialisation in Machine Learning and Computer Vision.
- Bachelor's thesis: "Development of Deep Learning Models Based on Recurrent Neural Networks for the Classification of Electrocardiograms."

2017 – 2022
Bachelor's Degree in Mathematics
University of Granada, Granada, Spain.
- Bachelor's thesis: "Development of Deep Learning Models Based on Recurrent Neural Networks for the Classification of Electrocardiograms."

Experience

2023 - present
PhD Student, Predoctoral Fellowship
Department of Computer Science and Artificial Intelligence, University of Granada, Granada, Spain.
- Funded by a research contract from the Regional Government of Andalusia.

2025
Visiting PhD Student
University of Nottingham, Nottingham, United Kingdom.
- Collaborated in the Computer Vision Lab under the supervision of Shreyank N. Gowda. We worked with diffusion models applied to Zero-Shot Learning.

2023 - 2025
Research Assistant, INCIBE Project
University of Granada, Granada, Spain.
- Research on General-Purpose Artificial Intelligence.

2023
Quantum Computing Researcher
Banco Bilbao Vizcaya Argentaria (BBVA), Madrid, Spain.
- Research in Quantum Computing and Quantum Machine Learning.

Research

2025 **Semantic-Inductive Attribute Selection for Zero-Shot Learning**
J. Herrera-Aranda, G. Gómez-Trenado, F. Herrera, and I. Triguero. Logic Journal of the IGPL.

2025 **A first approach to refine semantic spaces in zero-shot learning with a genetic algorithm**
J. J. Herrera, F. Herrera, and I. Triguero. IEEE Congress on Evolutionary Computation (CEC), pp. 1-4.

2024 **A preliminary study on preprocessing the semantic space in zero-shot learning**
J. J. Herrera Aranda, F. Herrera, and I. Triguero. International Conference on Hybrid Artificial Intelligence Systems (HAIS), pp. 177-189.

2024 **Aprendizaje Generalizado de Cero Ejemplos: Revisión del Estado del Arte y Estudio Preliminar de la Influencia del Preprocesamiento**
XI Simposio Teoría y Aplicaciones de Minería de Datos, Conferencia de la Asociación Española para la Inteligencia Artificial (CAEPIA).

2023 **Quantum Adversarial Generative Networks**
MSc thesis, University of Nebrija, Spain.

2022 **Development of Deep Learning Models Based on Recurrent Neural Networks for the Classification of Electrocardiograms**
BSc thesis, University of Granada, Spain.

Training courses

July 2026 **Agentic AI course**
University of Granada, Granada, Spain.

Feb 2026 **International Winter School on General-Purpose Artificial Intelligence**
University of Granada, Granada, Spain.

July 2024 **International Summer School on Data Science - Trustworthy, Safe and Governable AI**
AdiaLab, University of Granada, Granada, Spain.

Awards

Feb 2026 **Best Project, International Winter School on General-Purpose Artificial Intelligence**
University of Granada.

Dec 2024 **Best Project, II DaSCi Pre-Doctoral Workshop**
University of Granada.

July 2024 **Best Project, International Summer School on Data Science - Trustworthy, Safe and Governable AI**
AdiaLab, University of Granada.

Teaching

2026 - 2027 **Theory of Computation**
Bachelor's Degree in Computer Science and Mathematics, University of Granada, Granada, Spain.

2026 - 2027 **Introduction to Data Science**
Bachelor's Degree in Data Science and Artificial Intelligence, University of Granada, Granada, Spain.

2025 - 2026 **Artificial Intelligence**
Bachelor's Degree in Computer Science and Mathematics, University of Granada, Granada, Spain.

2025 - 2026 **Data Mining**
Bachelor's Degree in Statistics, University of Granada, Granada, Spain.

2025 - 2026 **Machine Learning, Teaching Assistant**
Bachelor's Degree in Computer Science, University of Granada, Granada, Spain.

Outreach and Talks

2026 **Three Minute Thesis Competition**
Competition organised by the Coimbra Group for PhD students; reached the final. Granada, Spain.

2026 **Poster Session: Zero-Shot Learning**
Winning final project in General Purpose AI Winter School. Granada, Spain.

2025 **Talk: "Zero-Shot Learning for Image Classification using Generative Models."**
Talk for the Computer Vision Lab during a visiting period at the University of Nottingham, Nottingham, United Kingdom.

2025 **Talk on the dangers of ChatGPT**
Pint of Science. Granada, Spain.

2025 **Poster presentation**
IEEE Congress on Evolutionary Computation. Hangzhou, China.

2024 **Talk: "Preprocessing the Semantic Space in Zero-Shot Learning."**
DaSCi Pre-Doctoral Workshops. Granada, Spain.

2024 **Mini-tutorials on using different AI tools**
European Researchers' Night. Granada, Spain.

2024 **Jailbreaking in Large Language Models.**
International Summer School on Data Science: Trustworthy, Safe and Governable AI. Granada, Spain.

News and Media

2025 **Canal UGR**
Reached the final of the Three Minute Thesis competition.

2024 **Radiolab UGR**
Invited speaker on a podcast as a PhD student in Artificial Intelligence.

Languages

Spanish	Native speaker
English	Fluent